

24V/12V BATTERY VOLTAGE Level Indicator

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Sometimes we do not notice that the battery has drained out and as a result the connected equipment does not work properly. By connecting this simple 24V/12V battery voltage level indicator circuit across the battery, you can check its voltage level any time.

The circuit can be used to check 24V and 12V batteries both. The battery level is indicated by the ten LEDs. Each glowing LED indicates 10% of battery voltage level. So, when LED1 through LED5 are glowing, or just LED5 is glowing, it means the battery is around 50% charged. The circuit can help in checking the batteries for a car, inverter, solar system, etc.

The author's circuit rigged on a breadboard is shown in Fig. 1 while the circuit's block diagram is shown in Fig. 2.

As shown in the block diagram, when LED1 alone glows it means the battery level is only 10%. With

LED2 also glowing it means the battery level is 20%, and so on. When all the LEDs (LED1 through LED10) are glowing, it means the battery level is 100% and it is fit for use.

Switch S1 is used to monitor either 24V or 12V battery. Switch S2 helps select dot mode or bar mode display. In dot mode only the last LED representing the voltage glows

while in bar mode made all the LEDs up to that LED glow. The choice is yours.

Circuit and working

Circuit diagram of the battery voltage level indicator is shown in Fig. 3. It is built around IC LM3914 dot/bar driver IC1, 1N4007 diodes D1 and D2, ten 5mm LEDs

(LED1 through LED10), 12.1V Zener diode (ZD1), BC547 transistor T1, BC557 transistor T2, presets VR1 and VR2, and piezo buzzer PZ1.

The LED display comprising LED1 through LED10 shows voltage level of the 24V or 12V battery selected. This display can be set in dot mode or bar mode. For showing the voltage level in bar mode, connect pin 9 of the IC to positive terminal of the battery using switch S2. For showing the voltage level in dot mode, simply open switch S2.

The 12.1V Zener diode ZD1 is used along with BC547 tran-

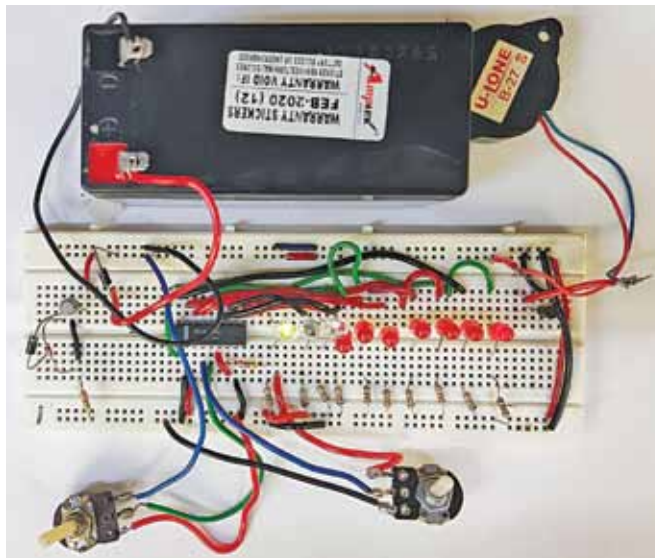


Fig. 1: Author's circuit on breadboard

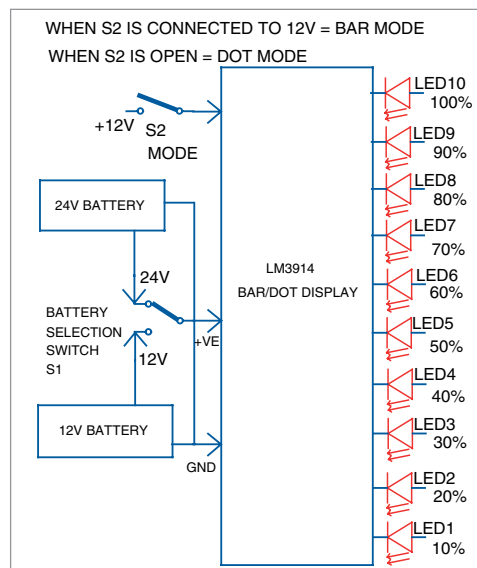


Fig. 2: Block diagram

PARTS LIST

Semiconductors:

IC1	- LM3914 dot/bar display driver IC
T1	- BC547 NPN transistor
T2	- BC557 PNP transistor
D1, D2	- 1N4007 rectifier diode
LED1-LED10	- 5mm LED
ZD1	- 12.1V Zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 3.3-kilo-ohm
R2	- 1.2-kilo-ohm
R3-R12	- 150-ohm
VR1	- 100-kilo-ohm preset
VR2	- 10-kilo-ohm preset

Miscellaneous:

S1	- SPDT switch
S2	- On/off switch
PZ1	- Piezo buzzer
BATT.1	- 24V battery
BATT.2	- 12V battery